

Introducing Asymflation: A Computational Framework for Macroeconomics

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Abstract

Asymflation is defined here as asymmetric inflation dynamics where prices rise rapidly in response to cost shocks but decline slowly or incompletely when costs fall, creating persistent upward pressure on the aggregate price level through downward price rigidities and behavioral asymmetries in price adjustment. This paper develops Asymflation as a computational framework for macroeconomics by linking the concept to asymmetric inflation dynamics under cost shocks, formalizing faster upward pass-through than downward adjustment in prices, and operationalizing Asymflation with a transparent index built from channels such as input cost shock, price increase speed, price decrease speed, and inflation expectations, while also showing how Asymflation can support comparative statics, empirical measurement, and policy simulation in reproducible economic research.

1 Introduction

Asymflation addresses a recurring problem in macroeconomics: analysts often observe asymmetric inflation dynamics under cost shocks, yet the mechanism is usually described with long phrasing that shifts across papers, sectors, and datasets. Asymflation provides a compact label without changing the underlying meaning. In practical research, the cost of inconsistent naming is real. If one paper describes the mechanism through pricing, another through governance, and a third through institutional friction, the literature can look

fragmented even when the same economic process is being studied. This paper treats Asymflation as a working term that should organize theory, code, and evidence at the same time.

The canonical definition of Asymflation is asymmetric inflation dynamics where prices rise rapidly in response to cost shocks but decline slowly or incompletely when costs fall, creating persistent upward pressure on the aggregate price level through downward price rigidities and behavioral asymmetries in price adjustment. Framed this way, Asymflation is neither a slogan nor a broad umbrella for unrelated cases. Asymflation identifies a specific pathway through which incentives, frictions, and measurable channels interact. That is why the paper places Asymflation explicitly inside Macroeconomics, Monetary Economics / Price Theory, and Behavioral Economics and keeps the classification close to JEL codes D40, E31, E52. The goal is to make Asymflation legible to researchers who want a reusable label without losing disciplinary specificity.

Asymflation also benefits from a stable linguistic identity. Portmanteau of "asymmetric" and "inflation." The term links asymmetric price adjustment dynamics with inflation behavior and parallels established forms such as "stagflation" and "disinflation.". Naming the mechanism clarifies what belongs inside the concept and what does not. The term is intended to survive outside a single article: it should still make sense in a memo, a replication package, a referee report, or a regulator's briefing note. The aim of this paper is therefore to move Asymflation from coined terminology into something that reads like an actual working concept in applied economics.

2 Conceptual Framework

Asymflation is best understood as a structured combination of channels rather than a single proxy. In the current implementation, the most salient channels include input cost shock, price increase speed, price decrease speed, and inflation expectations. These variables matter because each captures a different margin through which the mechanism becomes visible. Some channels amplify the underlying process, especially input cost shock, price increase speed, and inflation expectations. Other channels operate as offsets; in this package, the most important mitigating margin is price decrease speed. That interpretation makes Asymflation easier to audit than a loose narrative would be.

The conceptual contribution of Asymflation is to separate mechanism from symptom. Related concepts such as Rockets and feathers, Price rigidity, and Cost pass-through asymmetry often describe adjacent outcomes, institutional settings, or historical precedents. Asymflation, by contrast, isolates the specific process through which faster upward pass-through than downward adjustment in prices. This is valuable because policy responses differ depending on whether the analyst is observing the mechanism itself, a background condition that enables it, or a downstream effect that appears later.

Asymflation also highlights a distributional question. In many settings, the entity making the optimization choice captures flexibility, scale, or strategic advantage, while another party absorbs timing costs, information costs, or valuation uncertainty. That asymmetry gives Asymflation positive and normative relevance at the same time. The positive side concerns transmission and measurement; the normative side concerns burden-sharing, bargaining position, and the institutional margin on which reform is most likely to matter.

Concrete examples make the point easier to see. Asymflation is visible in sectoral pricing after energy and shipping shocks, in retail pass-through patterns during disinflation episodes, and in firm-level margin adjustments under sticky expectations. In each setting, the details differ, but Asymflation remains analytically coherent because the same broad mechanism is present and the same channel logic can be traced. A useful way to read the package is to ask which actors control the high-weight channels, which actors absorb the downside when those channels move, and what institutional lever would change the score.

2.1 Formal Definition

Asymflation is represented as a latent construct summarized by observable channels. Let $x_{k,t}$ denote the normalized level of channel k at observation t , and let w_k denote its contribution to the composite measure. The paper treats Asymflation as present when the weighted combination of channels indicates a persistent and economically meaningful realization of the underlying mechanism. The purpose of this representation is not to claim that Asymflation is exhausted by one formula. Instead, the formal definition gives Asymflation a baseline that can be replicated, stress-tested, and revised without losing contact with the canonical meaning.

$$T_t = \sum_{k=1}^K w_k \tilde{x}_{k,t}, \quad \sum_{k=1}^K w_k = 1 \quad (1)$$

$$\tilde{x}_{k,t} = \frac{x_{k,t} - \min(x_k)}{\max(x_k) - \min(x_k)} \quad (2)$$

$$W_t = Y_t - C_t(\tilde{x}_{1,t}, \dots, \tilde{x}_{K,t}) \quad (3)$$

3 Mathematical Formalization

The mathematical structure of Asymflation begins with a channel vector that links theory to observables. Each component corresponds to a distinct economic lever rather than an arbitrary feature transformation. That matters because Asymflation is supposed to support interpretation, not merely classification. When an amplifying channel increases, analysts should be able to explain why Asymflation becomes more intense; when a mitigating channel increases, they should be able to explain why Asymflation weakens. In this sense, Asymflation is designed for comparative statics as much as for scoring.

The next step is to connect Asymflation to incentives. A general decision-maker chooses an action or institutional rule that responds to state variables while balancing efficiency against the frictions implied by the mechanism. When those incentives systematically reinforce faster upward pass-through than downward adjustment in prices, the score rises. When institutions weaken the pass-through from incentives to outcomes, the score falls. That logic is flexible enough to accommodate organizational settings, market-wide equilibria, and platform-mediated interactions, which is why Asymflation can be studied across several domains without losing its identity.

Asymflation also admits a threshold interpretation that is useful in practice. Because the score is normalized and weighted, a move from one qualitative band to another has a substantive meaning: the mechanism has become materially stronger or weaker rather than merely noisier. For policy users, that feature makes Asymflation appropriate for triage workflows, compliance reviews, and institutional benchmarking. For researchers, it makes Asymflation useful in event studies and pre/post comparisons where the question is whether the mechanism intensified after a rule change.

A further advantage of the formalization is that the concept can be decomposed. Analysts can inspect whether changes in the score are driven by a single dominant margin or by several channels moving together. That is important in applied work because interventions that improve one channel may leave the rest of the mechanism intact. Reporting both the index and its component contributions keeps Asymflation interpretable when it is used in panel data, policy dashboards, or comparative case studies.

Finally, the formalization emphasizes that this mechanism is dynamic. The intensity of the construct depends on the joint evolution of channels, weights, and institutional responses. A concept that cannot be moved by policy or design changes is not very useful in applied work. The measure, by contrast, is explicitly defined so that changes in identified channels map into predictable changes in the score.

$$X_t = (\text{input_cost_shock}_t, \text{price_increase_speed}_t, \text{price_decrease_speed}_t, \text{inflation_expectations}_t, \text{market_power}_t) \quad (4)$$

$$T_t = \sum_{j \in P} w_j \tilde{x}_{j,t} + \sum_{m \in N} w_m (1 - \tilde{x}_{m,t}) \quad (5)$$

$$\mathcal{L}_t = \mathbb{E} [\phi_0 + \phi_1 q_t - \phi_2 s_t + \phi_3 z_t] \quad (6)$$

$$\theta_{t+1} = \theta_t + \eta \nabla_{\theta} \mathbb{E} [\mathcal{L}_t \mid X_t] \quad (7)$$

$$\Delta T \approx \sum_{j \in P} w_j \Delta \tilde{x}_j - \sum_{m \in N} w_m \Delta \tilde{x}_m \quad (8)$$

Proposition 1 (Monotonicity of Asymflation). If all weights are nonnegative and sum to one, then the score is increasing in positively signed channels and decreasing in negatively signed channels.

Proof sketch. Differentiate equation (5) channel by channel. Because the score is linear in normalized channels, marginal effects equal the associated weights. Positive and negative signs follow directly from whether the channel enters directly or through its complement.

Lemma 1 (Policy transmission). Any intervention that raises a mitigating channel or lowers an amplifying channel weakly reduces the steady-state score whenever other channels are held fixed.

Proof sketch. Apply equation (8) to a local policy perturbation. The direction of the score change is determined by the sign of the relevant weight and by the sign convention assigned to the channel in the index.

4 Computational Implementation

The computational implementation of the concept is designed to look like something a researcher could actually use. The module validates expected columns, normalizes each channel to the unit interval, computes the weighted score, and classifies observations into qualitative bands. That design choice matters because this mechanism should remain inspectable from top to bottom. A user can see how the construct is defined, how the measure is computed, and how a policy counterfactual changes the score without reading hidden preprocessing layers.

The package also enforces a minimal reproducibility discipline. A caller has to provide the expected channels, the weights must sum to one, and the simulation interface makes the counterfactual question explicit rather than implicit. This does not turn the construct into a full structural model, but it does make the object usable in a notebook, a classroom demonstration, or a lightweight empirical pipeline. For a coined term, that difference matters: the measure looks less like a slogan once it can be loaded, calculated, and perturbed in code.

The bundled synthetic dataset supports end-to-end testing of the measure. Researchers can trace how the channels combine into a score, inspect the distribution of the concept, and run policy scenarios that reduce one channel at a time. In that sense, the package does more than name the concept. It provides a plausible workflow for taking this mechanism from definition to data, from data to code, and from code to empirical interpretation.

```
import pandas as pd
from asymflation import AsymflationCalculator

df = pd.read_csv("asymflation_dataset.csv")
calc = AsymflationCalculator()
out = calc.calculate_asymflation(df)
```

5 Applications and Examples

Asymflation is well suited to applied diagnostics in contexts where asymmetric inflation dynamics under cost shocks. One use of Asymflation is cross-sectional comparison: analysts can compare units exposed to the same broad environment but differing institutional detail, and then ask whether the variation in the score comes from governance, incentives, technology, or frictions. That keeps the analysis from collapsing into broad narrative claims.

A second use of the concept is scenario analysis. Because the score is built from explicit channels, reforms can be mapped into expected changes in this mechanism. In this domain the most plausible levers are competition policy, anti-gouging enforcement, expectation management, and targeted cost-relief interventions. The practical value of the construct is that each lever can be translated into a channel-level hypothesis rather than a vague expectation that conditions might improve.

A third use of the concept is communication. Working papers, memos, and governance dashboards often need a short label that means the same thing across tables and contexts. This mechanism fills that role without sacrificing analytic discipline. The examples most relevant to the construct include sectoral pricing after energy and shipping shocks, retail pass-through patterns during disinflation episodes, and firm-level margin adjustments under sticky expectations. Those are not identical cases, but they are close enough for the measure to improve comparability rather than create confusion.

Applied users can also read the construct as a screening device. A high value does not by itself settle causality or welfare, but it does indicate where deeper institutional investigation should begin. In that role, the measure sits comfortably between a descriptive dashboard metric and a full-blown structural estimate.

6 Empirical Considerations

Empirical work on the concept begins with careful channel construction. Observed proxies should match the canonical definition rather than convenience alone. In practice, that means documenting how each variable maps onto the mechanism, how it is normalized, and why its direction is positive or negative in the score. For the current package, representative measure-

ment choices would look like the following: input cost shock can be proxied with a documented normalized indicator drawn from administrative, survey, or transaction data; price increase speed can be proxied with pass-through elasticities or event-window adjustment rates; price decrease speed can be proxied with pass-through elasticities or event-window adjustment rates; inflation expectations can be proxied with survey expectations or text-based sentiment indices. This discipline matters because this mechanism is intended to support comparison across settings. If the channels drift too far from the definition, the term survives but the concept does not.

Data construction is equally important. A serious empirical study of the measure would not rely on a single spreadsheet. Instead, researchers would typically combine scanner data, producer and consumer price microdata, earnings-call text, and cost exposure measures derived from input-output tables. That kind of source triangulation matters because the mechanism is rarely visible in one table alone. The score becomes more credible when institutional metadata, behavioral traces, and outcome measures line up.

Identification is a separate challenge. Common shocks, institutional transitions, and measurement error can all mimic the mechanism captured by this mechanism. Researchers should therefore combine the score with event studies, staggered adoption designs, discontinuities, placebo tests, or cross-sectional robustness checks where possible. The score is best understood as a structured diagnostic layer, not as a complete causal estimator. That division of labor improves empirical clarity because it lets the construct organize evidence without claiming more than the data can support.

External validity should also be treated carefully. A value of the concept estimated in one institutional environment does not automatically travel to another because contracts, governance, market design, and enforcement differ. In the present domain, the key empirical difficulty is separating true asymmetric adjustment from common shocks, composition effects, and changes in product mix. That is why the package is framed as a transparent baseline and not as a universal calibration.

Asymflation also encourages metadata discipline. Knowing when a system changed, who could override a rule, how a benchmark was constructed, or when a regulation took effect can materially alter interpretation. The repository therefore treats code, data, and documentation as one package. That package structure makes Asymflation easier to revisit later, which is part of what makes the current package look like a real working term rather than a placeholder.

Asymflation is also easier to defend in review when authors can show exactly which channel moved, what data generated that movement, and which institutional feature plausibly explains it.

That practical auditability is one reason the measure can be useful in interdisciplinary settings where economists, lawyers, policy staff, and engineers need a common reference point.

A term like the construct earns its place only if it shortens communication while preserving the mechanism, and the package is structured to make that tradeoff visible.

In that sense, this mechanism is meant to function as a durable label for a recurring empirical pattern rather than as an exercise in vocabulary for its own sake.

The value of the concept is strongest when the same label can survive contact with data cleaning, model specification, sensitivity checks, and policy discussion.

For empirical teams, the measure is especially useful when data engineering, model estimation, and policy interpretation are handled by different people who still need a shared vocabulary for the same mechanism.

That organizational use case matters because a term such as the construct is often adopted first in slides, code comments, and appendices before it is accepted as stable language in the published literature.

Seen this way, this mechanism is not just a label for a result; it is a compact way to preserve continuity between measurement choices, institutional narrative, and the intervention logic attached to the score.

Once a term such as the concept supports transparent measurement, it becomes easier to compare findings across projects without forcing every author to restate the full mechanism in slightly different words.

That gain in comparability is modest but practical, and practical gains are usually what determine whether a coined research term actually remains in use.

7 Discussion and Extensions

Asymflation has limits. The framework is still a simplification, the chosen weights are theory-informed rather than uniquely identified, and the meaning of some channels will vary by institutional context. Yet those limits are not arguments against using Asymflation. They are reasons to use Asymflation

carefully. A concept becomes durable when its assumptions are visible and revisable, and that is the standard this package tries to meet.

Future work can extend the concept in several directions. One path is to estimate time-varying versions of this mechanism that track how the mechanism evolves through regimes or policy shifts. Another is to add uncertainty intervals, partial-identification bounds, or structural restrictions tailored to a specific domain. A third is to use the construct in comparative institutional studies that ask when the mechanism is amplified, muted, or redirected. Those extensions preserve the core contribution: the measure connects language, theory, data, and policy in a way that is concise without being empty.

That is also why additional detail matters. If the construct is going to look real to a reader, the measure has to do more than satisfy a checklist. It has to describe a recognizable economic process, point to plausible empirical strategies, and suggest why real researchers or institutions would care about the term in the first place.

8 Conclusion

Asymflation gives researchers a concise but disciplined way to study asymmetric inflation dynamics under cost shocks. By anchoring the term in the canonical definition, formalizing the mechanism with explicit equations, and implementing Asymflation in reproducible code, the paper turns the phrase into an operational research object rather than a loose descriptor.

The practical payoff is immediate. The concept can be used in empirical pipelines, policy design, audit workflows, and teaching materials without severing the link between concept and measurement. That combination of clarity, portability, and applied usefulness is the reason to present this mechanism as a real working term and not merely a coined label.

Code and Data Availability

All replication materials are available at: <https://github.com/EconomicTermDevelopments/asymflation-economics> and <https://huggingface.co/datasets/EconomicTermDevelopments/asymflation-economics>

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